Iterative List Decoding

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Overview

1. Introduction to Iterative List Decoding

2. Error Patterns and Syndromes
   a) Parity Check Matrix Construction
   b) Distribution And Properties of Syndrome
   c) Properties, Lemma, Theorems for BF Algorithm

3. Iterative List Decoding

4. Codes from Finite Geometries
   a) Minimum Weights in $C(s)$ and $C(\hat{s})$

Definition

- binary code (n, k, d)
- Syndrome $s = Hy^{'}$ with weight u
- H is a parity check and syndromes of columns have the same weight γ
- y is a received vector of length n

Decoding Algorithm Overview (non-list version)

- decoding overview
  - plug in y to get s
  - Based on extended syndrome, flip the bit that reduces weight of extended syndrome u
  - repeat 1 and 2 until $u \rightarrow 0$

The algorithm used for list decoding is a variation of the above with a tree-structured search. Whenever there is a choice between several bits, all possibilities are tried in succession.

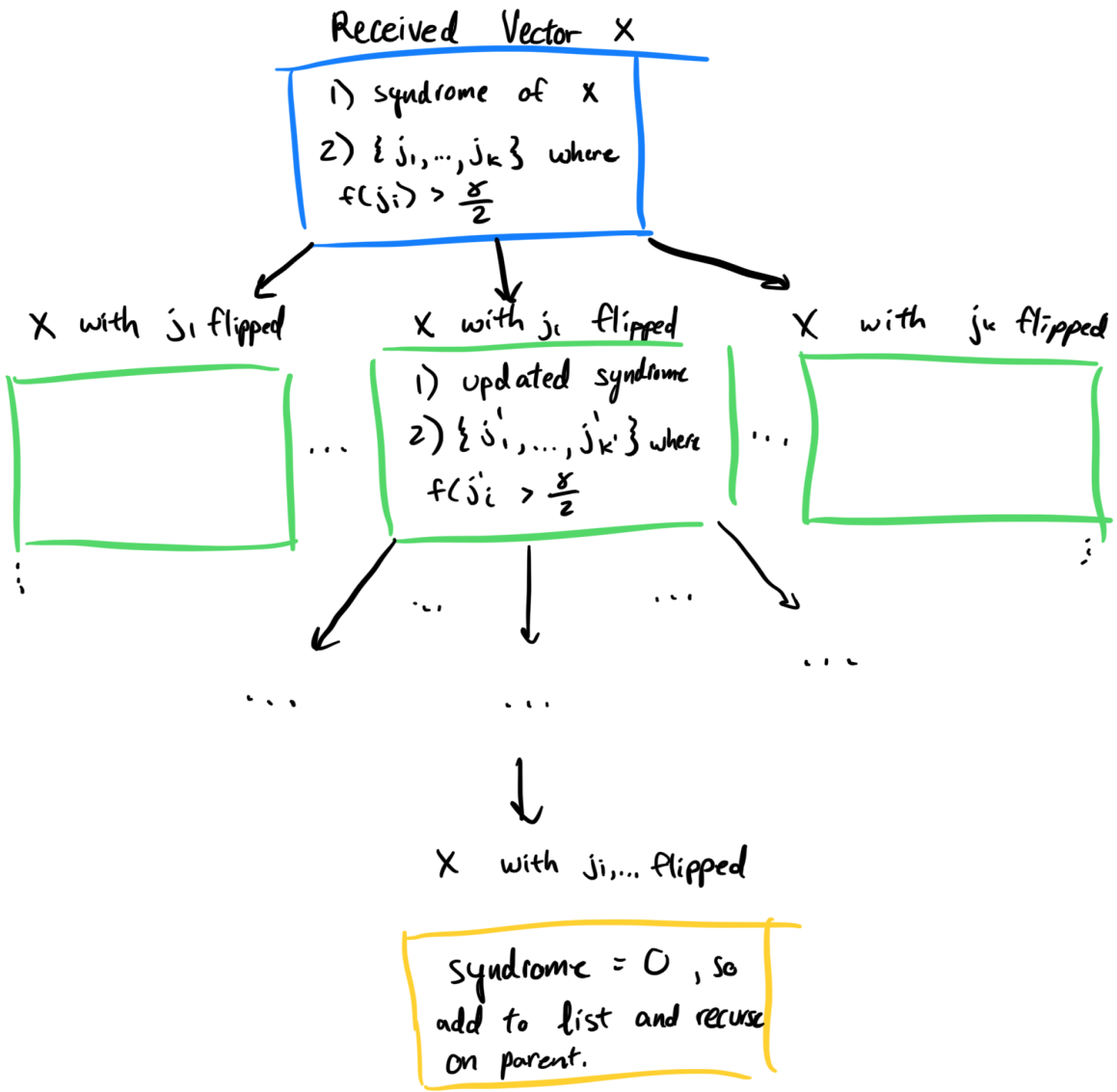
List Decoding Algorithm Overview

- Compute the syndrome from parity check matrix H and and extended syndrome Hy'
- Apply the function $f(j)$ to calculate violations for each position j
- Keep an ordered list of enteries where $f(j) > \frac{\gamma}{2}$
- Flip a bit in the ordered list and repeat from step 2 until $u = 0$ or $u > 0$ and no further flips can be made to decrease u. If $u = 0$, add the final codeword to the list.
- Backtrack to the first node where a different bit could have been flipped and repeat until all nodes are explored.

List Decoding Algorithm Observations

- We can also make this algorithm more efficient by either increasing the threshold of $\frac{\gamma}{2}$ or flipping the high values of $f(j)$ first

- You can imagine that the first algorithm is simply one path in the decision tree from the list-decoding algorithm.


Received Vector x
1) syndrome of x
2) $\{j_1, \ldots, j_k\}$ where $f(j_i) > \frac{\delta}{2}$
x with j_1 flipped
x with j_i flipped
x with j_k flipped
1) updated syndrome
2) $\{j'_1, \ldots, j'_{k'}\}$ where $f(j'_i) > \frac{\delta}{2}$
x with $j_i, \ldots$ flipped
syndrome = 0, so add to list and recurse on parent.

Parity Check Matrix Construction

- Regular Parity Check Matrix H_{mxn}
  - syndrome of each column has low weight γ

  - For any pair of columns c_i and $c_j (j \neq i)$, there is at most one row where they both have one's. (Note this also implies the same must hold for rows)

Important Theorems/Lemmas/Properties

Property 5	Minimum distance of code $d \geq \gamma + 1$
Lemma 1	A unique closest codeword is fond by the BF algorithm if $\omega \leq \frac{\gamma+1}{2}$
Lemma 2	If $f(j) > \omega$, any error pattern of weight ω with that syndrome must include position j
Lemma 3	$\sum_{j \in J} f(j) = \sum_{i \text{ odd}} ib_i$
Lemma 5	if $u > w\lfloor \frac{\gamma}{2} \rfloor$, then list decoding could flip at least one error position
Theorem1	If $u > w^2$, the errors are corrected by the BF algorithm and there is a unique coset leader.
Theorem2	Let $F = \{J_i\}$, F is decoded by the list BF algorithm if and only if for every error pattern $J_i \in F$ there is at least one position such that $f(j) > 2$

Property 5

Based on the construction of our parity-check matrix H, the following inequality holds for the syndrome of weight u of an error vector with weight w:

$$\omega\gamma - \omega(\omega - 1) \leq u \leq \omega\gamma$$

Since the above holds for any vector with weight w, a non-zero codeword (weight > 1) has $u = 0$ and thus $w \geq \gamma + 1$. Since this is a linear code, the minimum distance is the lowest weight codeword $\rightarrow \mathbf{d} \geq \gamma + \mathbf{1}$.

Lemma 1

A unique closest codeword is found by the BF algorithm if $w \leq \frac{\gamma+1}{2}$ for an error weight w.

This can be seen by considering the variable/check node graph and by the property of H that any two columns share at most one index where they both have a 1. Full proof provided in separate document.

Definition

Given a parity-check matrix H and a syndrome s, We define $f(j)$ to be the number of rows that give a 1 in the syndrome due to the contribution index j of an input vector.
This is equivalent to the number of unsatisfied check nodes connected to the jth variable node in the Tanner graph. (note $f(j) \leq \gamma$)

Example: Consider the parity check matrix

$$H = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}$$

Given the syndrome (1 0),
$f(0) = f(1) = s[0] = 1$ and $f(2) = f(3) = s[1] = 0$.

Lemma 2

If for a given syndrome and some position $j, f(j) > \omega$, any error pattern of weight w with that syndrome must include position j.

Due to the property that each pair of rows r_i, r_j in H must share a 1 in at most one index, a correct bit position can have at most w positions where it screws up the parity.There are are ω errors in the received vector and due to the property that each pair of rows r_i, r_j in H must share a 1 in at most one index, v_j can be connected to at most ω faulty parity checks if it's correct. Hence, if $f(j) > \omega$ then v_j is an error.

Definition 2

Let us refer to a variable node as a point, and the set of points $v_1, ..., v_\gamma$ where a row of H has 1s as a line. By construction of H:

- a pair of points (x_1, x_2) is present in at most one line
- the intersection of two distinct lines is at most a single point

Define $b(j)$ as the number of errors on line j.
Define b_i as the number of lines that have exactly i errors.

Definition 2 (slide 2)

Define $b(j)$ as the number of errors on line j.
Define b_i as the number of lines that have exactly i errors.

- The total number of edges going out of errors

$$\gamma\omega = \sum_i i b_i$$

- the syndrome weight

$$u = \sum_{i \text{ odd}} b_i$$

Lemma 3

Denote the set of indices of errors with J. Then we have

$$\sum_{j \in J} f(j) = \sum_{i \text{ odd}} i b_i$$

Both sums give the total number of edges going out from errors to unsatisfied checks.

Theorem 1

If $u > w^2$, the errors are corrected by the BF algorithm and there is a unique coset leader.

Assume that $u > w^2$. If $2\omega \leq \gamma$ then the conclusion directly follows from Lemma 1. Assume that $2\omega > \gamma$. From Lemma 3 we have:

$$\sum_{j \in J} f(j) = \sum_{i \text{ odd}} i b_i \geq \sum_{i \text{ odd}} b_i = u > \omega^2$$

Notice that $|J| = \omega$, so $\sum_{j \in J} f(j)$ is a sum of ω terms and it is strictly greater than ω^2, therefore there must be at least one term that is strictly greater than ω. Let $f(i) > \omega > \frac{\gamma}{2}$, then by Lemma 2 v_i is an error and will be flipped by the algorithm.

Theorem 1 continued

Flipping v_i results in a new syndrome of weight:

$$u' \geq u - f(i) \geq u - \gamma \geq \omega^2 + 1 - \gamma = (\omega - 1)^2 + 2\omega - \gamma > (\omega - 1)^2$$

where the last inequality follows from our assumption $2\omega > \gamma$. Note that $\omega - 1$ is the weight of the error pattern after flipping v_i and hence the conclusion is reached by repeating this argument.

Theorem 2

Let $F = \{J_i\}$, be a set of error patterns such that any subset of J is also in F. F is decoded by the list BF algorithm if and only if for every error pattern $J_i \in F$ there is at least one position such that $f(j) > \frac{\gamma}{2}$

This theorem basically tells us that as long as each error pattern in F has at least one position j such that it satisfies the clear condition in section II that it is an error position, our tree-like algorithm structure will eventually uncover all error-patterns in this set F that we are searching for.

Lemma 5

For any error pattern of weight w in a coset with syndrome weight u, the list decoding algorithm will flip at least one error position if

$$u > w\lfloor\frac{\gamma}{2}\rfloor \tag{3.1}$$

Since s is the sum of ω columns of H, one has more than or equal to $\frac{u}{\omega}$ ones. Hence, there exists an error position $\hat{j}$ with $f(\hat{j}) \geq \frac{u}{\omega} > \lfloor\frac{\gamma}{2}\rfloor$

Codes from Finite Geometries

- Now what if we were to think of decoding geometrically? (i.e. in lines vs. points)
- Let PG$(2, q)$ denote the projective plane over a finite field with alphabet size 2 (binary) with dimensions q $= 2^s$ (We can think of the projective plane as a consolidated version of our original plane)
- Let EH$(2, q)$ denote the Euclidean plane over a finite field with same dimensions as EG$(2, q)$
- Let H be the incidence matrix of lines $L_1, L_2, ..., L_b$ and points $P_1, P_2, .., P_n$ versus points or:

$$H = (h_{uv}) \text{ where } = \begin{cases} 1, & \text{if } P_v \in L_u \\ 0, & \text{otherwise} \end{cases} \tag{4.1}$$

Minimum Weights in $C(s)$ and $C(\hat{s})$

Lemma 6: Consider a finite geometry code $PG(2, q)$ with a cyclic H. Let c be a codeword of minimum weight. The set O of points corresponding to the nonzero positions satisfy:

1. any two points lie on a unique line
2. any line has either 2 or 0 points in common with O

and if we have a set of $\gamma + 1$ points satisfying 1 and 2, the incidence vector corresponds to a minimum-weight codeword.

- From Lemma 6, since for any line we either share 2 or 0 points, by definition we have an arc
- Specifically, for the set of points that correspond to the minimum weight codeword in $C(s)$, we have a $q + 2$ arc in $PG(2, q)$
- This arc will be vital in proving Theorem 3 on the next slide

Theorem 3 The number of minimum-weight words in

$$
\begin{array}{lll}
C(2) & \text{is} & 168 \\
C(3) & \text{is} & 32{,}704 \\
C(4) & \text{is} & 119{,}867{,}920 \\
\hat{C}(2) & \text{is} & 18 \\
\hat{C}(3) & \text{is} & 1{,}960 \\
\hat{C}(4) & \text{is} & 3{,}704{,}400
\end{array}
$$

5. Complete ML Decoding of the (73,45,10) PG Code

Note: Geometry is $PG(2,8)$, $q = 8$, $s = 3$. $\gamma = q + 1$, every line has 9 points.

Motivation: To analyze the cosets and coset leaders of this finite geometry code to show that either Lemma 5 or Theorem 1 holds, and therefore the BF algorithm can find a unique coset leader or a list of coset leaders.

Example: The all ones syndrome indicates a coset $(73, 9)$, and can be obtained by 73 error patterns of weight 9 (all coset leaders), where the positions are one of the 73 lines.

5. Complete ML Decoding of the (73,45,10) PG Code

A useful property: Given a (u, ω) coset, we can find vectors in the corresponding $(73 - u, \omega')$ coset by adding one of the weight 9 error patterns. We can minimize the weight of the new error pattern by choosing the line that has the greatest number of 1s, call it v, and changing it to $9 - v$.

Example: There are 73 cosets with single errors $(u, \omega) = (9, 1)$. Each coset also contains some error patterns of weight 9, which can be found by removing one 1 from a minimum weight codeword. Therefore, the complementary cosets are $(64, 8)$ and they each have a list size of 9 (each point is included in 9 min weight codewords).

5. Complete ML Decoding of the (73,45,10) PG Code

Continuing this line of analysis, some of the cosets are found to be:

$(9,1)$	$(16,2)$	$(21,3)$
$(25,3)$	$(32,4)$	$(28,4)$
$(24,4)$	$(24,6)$	$(41,5)$
$(33,5)$	$(29,5)$	$(25,5)$
$(48,6)$	$(57,7)$	$(64,8)$
$(73,9)$		

A modification to the algorithm: It is found that each error point in $(24,6)$ has 4 parity failures. So the algorithm would stop in this scenario. We can flip one of the positions to get to a $(25,5)$ coset in this case. Similarly, flipping a bit in $(25,5)$ coset leads to either $(24,4)$ or $(24,6)$. If $(24,6)$ is reached the algorithm should backtrack.

5. Complete ML Decoding of the (73,45,10) PG Code

Lemma 7: For cosets of min weight $\omega = 8$, u is at least 36.

Theorem 4: The modified BF algorithm is complete and ML. The list includes all coset leaders.

References

1. J. Justesen, T. Hoholdt and J. Hjaltason, "Iterative List Decoding of Some LDPC Codes," https://ieeexplore.ieee.org/abstract/document/4373410

THANK YOU